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SUBJECT: Supplemental Context and Theoretical Foundation for PRD Manuscript es2026jan27_967

To the Editorial Office of Physical Review D,

I am submitting the attached PDF, *Distributed Field Processing V: Field Closure and the Topology of Gravity*, as foundational context for the topological derivation presented in my primary manuscript.

As an engineer with 20 years of experience in technical systems and data science, I approached the Hubble Tension as a field topology problem built upon the hypothesis that Tetrad topology is fundamental to the tension. Recently, I began compiling my collection of research into formal documents and have posted the series on Zenodo. While these papers are newly public, they represent a long term investigation into field closure and field settlement as mechanical processes analogous to engine mechanics, yet maintained in strict alignment with textbook Tetrad thinking. Paper V explicitly demonstrates how $k = 4$ drives the organization of the field topology and its mechanisms.

These topological mechanisms are well established in textbook gravity theory. The $k = 4$ structure is required to close the local frame in Tetrad gravity (1 timelike + 3 spacelike basis vectors). It also serves as the fundamental quantum of space in Loop Quantum Gravity (LQG), where 4-valent spin network nodes correspond to the faces of a quantum tetrahedron.

> **Key Foundational Context in Paper V:**
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> - **Topological Drivers (Sec. 2.6 & 4.2):** Establishes the Simplex Theorem as the geometric driver for $k = 4$ enclosure, providing the structural basis for the coupling channels used in my results.
> - **Field Settlement Mechanisms (Sec. 3.3 & 4.4):** Reinterprets static closure constraints as an active settlement process (READ-SETTLE) through which field configurations achieve consistency.
> - **Structural Bridge (Sec. 7.3):** Synthesizes these mechanical principles with textbook tetrad and LQG formalisms, showing that the Port Ratio ($\sqrt{k/\pi}$) is a natural consequence of $k = 4$ topology.

> - **Phenomenological Interpretations (Sec. 5.5):** Presents the Port Ratio as a candidate physical mechanism. Maintaining academic rigor, I offer three interpretations (Aperture, Periodic Mode-Constraint, and Topological Normalization) consistent with the 0.11% match to GW170817.

I believe this mechanical derivation provides the structural logic necessary to evaluate my resolution of the Hubble Tension. You may access the supporting series here:
Zenodo DOI: `10.5281/zenodo.18419531`

Sincerely,

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